

KLE TECHNOLOGICAL UNIVERSITY

Hubballi, Karnataka - 580 031



Dr. M. S. Sheshgiri Campus, Belagavi

A Senior Design Project Report on

WiFi Talking Robot using ESP8266

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Engineering

in

Electronics and Communication Engineering

by

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Semester VII

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CERTIFICATE

This is to certify that the project entitled “**WiFi Talking Robot using ESP8266**” is a bonafide work carried out by the student team of **Danamma Navarangi (02FE21BEC114), Ashwini M Dambal (02FE22BEC406), BibiAyesha Fanibandh (02FE22BEC408), Roopali Karadiguddi (02FE22BEC417)**. The project report has been approved as it satisfies the requirements with respect to the Senior design project work prescribed by the university curriculum for the award of the degree of **Bachelor of Engineering in Electronics and Communication Engineering (VII Semester)** in the Department of Electronics and Communication Engineering of KLE Technological University, Hubballi, for the academic year 2024-2025.

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DECLARATION

We hereby declare that the **Senior Design Project (SDP)** presented in this report, entitled ” **WiFi Talking Robot using ESP8266**”, submitted to KLE Technological University for the completion of the **Senior Design Project (Code: 20EECW401)** in the 7th Semester, is the original work carried out by us in the **Department of Electronics and Communication Engineering, KLE Technological University, Dr. M S Sheshgiri Campus, Belagavi**, under the guidance of **Prof. Anitha S Shirol, Designation of Staff, Department of Electronics and Communication Engineering**.

We further declare that, to the best of our knowledge and belief, the work reported herein has not been submitted as part of any other project or for the award of any course, degree, or diploma at this or any other university or institution. The results presented in this report are solely the outcome of our efforts.

We also confirm that all the work documented in this report has been completed by us.

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ABSTRACT

The project aspires to develop a WiFi-connected, voice-interactive robot that combines the principles of IoT, voice AI, and robotics. Using the ESP8266 module for internet connectivity, the robot will leverage Google Assistant to process voice commands, providing dynamic interaction and intelligent responses. Voice inputs will be transmitted to the robot, which will then respond audibly through a speaker powered by an amplifier, making it capable of answering questions in real-time.

Locomotion is achieved through the integration of four DC motors, enabling movement in all directions. These motors are controlled via relays, ensuring precise and efficient operation. To power the entire system, a robust 15V battery, coupled with a step-up transformer, provides the necessary voltage and stability for seamless performance.

The inclusion of a Bluetooth module further enhances the robot's functionality by enabling manual control, adding versatility to its operations. This dual control system demonstrates the flexibility of the robot in adapting to various use cases, whether autonomous or user-guided.

This project showcases the potential of integrating IoT and AI-driven voice technology into robotics. Beyond being a practical implementation, it serves as an educational platform to explore cutting-edge technologies, emphasizing their relevance in modern automation and interactive systems.

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Chapter 1

Introduction

Robots are rapidly becoming an integral part of modern life, finding applications in diverse areas such as industrial automation, healthcare, education, and even personal assistance. These intelligent machines are revolutionizing the way we interact with technology, providing solutions that are efficient, reliable, and user-friendly. This project aims to push the boundaries of robotics by combining Internet of Things (IoT) and Artificial Intelligence (AI) technologies to create a sophisticated robot capable of voice-based interaction and autonomous movement. At the heart of the project lies the ESP8266 microcontroller, a versatile and powerful module that enables the robot to connect to the internet. Through this connection, the robot can process natural language queries with the help of Google Assistant, allowing it to understand and respond to user inputs intelligently. This integration of voice AI elevates the user experience, making the robot not just a mechanical device but a conversational companion. In addition to its interactive capabilities, the robot is equipped with locomotion features, enabling it to navigate its surroundings and perform tasks autonomously. Four DC motors controlled by a relay-based circuit provide seamless movement in all directions, making the robot adaptable to various environments. The mobility aspect enhances its utility, making it suitable for applications such as educational demonstrations, household assistance, and interactive learning.

To ensure uninterrupted operation, the robot is powered by a robust 15V battery, which is further supported by a step-up transformer to maintain consistent voltage levels. For audio output, an amplifier is used to drive a speaker, ensuring clear and audible communication. This enables the robot to effectively answer questions and interact with users in real-time. Flexibility in control is another highlight of the project, achieved through the integration of a Bluetooth module. This feature allows users to manually control the robot, providing an additional layer of functionality and versatility. Whether operating autonomously or under user guidance, the robot is designed to perform efficiently in a variety of scenarios.

This project serves as a demonstration of the potential for integrating IoT, voice AI, and robotics into a cohesive system. Beyond its practical applications, it stands as an educational tool that showcases the possibilities of modern technology, inspiring innovation and learning in fields like robotics, automation, and artificial intelligence.

1.1 Motivation

1. Integration of Emerging Technologies:

- The project combines IoT, robotics, and AI, which are cutting-edge fields shaping the future of technology.
- Provides a hands-on opportunity to work with modern tools like ESP8266, voice assistants, and automation systems.

2. Real-World Problem Solving:

Addressing the need for interactive, intelligent, and mobile robots that can assist in daily tasks.

3. Education and Skill Development:

Helps in building essential skills in programming, electronics, and AI integration.

4. Personalized User Experience:

Motivation to create systems that interact naturally with humans, making technology more accessible.

5. Expanding Robot Applications:

Encourages the development of robots that are not only functional but also interactive and versatile.

1.2 Objectives

- 1. Voice-Based Interaction:** Enable the robot to process natural language queries using Google Assistant and provide real-time audible responses for intuitive human-machine interaction.
- 2. Autonomous Locomotion:** Implement a relay-controlled motor system to allow the robot to move in all directions, enabling it to navigate its environment efficiently.
- 3. Reliable Power Supply:** Utilize a 15V battery with a step-up transformer to ensure a stable power source capable of supporting all components, including motors, microcontrollers, and audio systems.
- 4. Audio Communication:** Integrate an amplifier and speaker system to deliver clear and audible responses, enhancing user engagement.
- 5. Flexible Control Options:** Incorporate a Bluetooth module to facilitate manual control of the robot, allowing users to operate it in different modes based on the application.
- 6. Integration of IoT and AI:** Leverage the ESP8266 module to connect the robot to the internet, showcasing the potential of IoT and AI in creating intelligent, interactive systems.
- 7. Educational and Practical Application:** Provide a platform to explore advanced technologies in robotics, automation, and voice AI, demonstrating their relevance for real-world applications in education, assistance, and entertainment.

1.3 Literature survey

1. Title: Wi-Fi Surveillance Bot with Real Time Audio and Video Streaming Through Android Mobile

Objective: The project focuses on integrating live video streaming, image capture, and storage capabilities with Wi-Fi control for seamless operation. It also includes real-time audio transfer to enable efficient communication and interaction.

Methods/Algorithms used:

- Arduino Uno R3
- NodeMcu ESP8266 Wi-Fi Module
- Android phones

Limitations:

- Limited Wi-Fi range
- Smartphone dependence
- Security risks

2. Title: Wireless Voice Control Robot

Objective: The project involves creating a voice-controlled robot that utilizes advanced speech-to-text conversion for interpreting user commands. A user-friendly Android app will be developed to provide an intuitive interface for controlling the robot.

Methods/Algorithms used:

- Arduino microcontroller.
- DC geared motors
- L293D motor driver.

Limitations:

- Limited range.
- Speech recognition errors.
- Bluetooth interference.

3. Title: Wifi talking robot

Objective: The project emphasizes enhancing human-machine interaction by providing remote control capabilities through Wi-Fi, enabling users to operate the system from a distance. Additionally, it incorporates task automation to improve efficiency and reduce manual intervention, making the system more reliable and user-friendly for a variety of applications.

Methods/Algorithms used:

- Speech Recognition: TTS
- Wi-Fi Communication.
- Motor Control.
- Obstacle Detection

Limitations:

- Short battery life.
- Dependence on stable Wi-Fi.
- Data privacy and security risks.



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Figure 1.1: Logo of KLE Technological University

1.4 Problem statement

To design and impliment a WiFi-controlled talking robot using ESP8266.

1.5 Application in Societal Context

- 1. Healthcare Assistance:** Assist elderly or disabled individuals by answering queries, reminding them of medication schedules, or offering companionship.
- 2. Retail and Customer Service:** Serve as an interactive guide in stores or malls, helping customers locate products or answer frequently asked questions.
- 3. Classroom Aid:** Act as a teaching assistant by providing answers to student queries, conducting quizzes, or explaining concepts.
- 4. Tour Guides:** Provide voice-guided tours in museums, exhibitions, or tourist spots, answering visitor queries about exhibits or landmarks.
- 5. Event Hosting:** Act as an MC or interactive host for events, responding to audience interactions and guiding the event flow.
- 6. Delivery Robot:** With slight modifications, it can be used for short-distance delivery tasks like carrying small items or packages.
- 7. Emergency Response:** Provide vital information during emergencies or guide individuals to safety in situations like fires or evacuations.
- 8. Gaming and Entertainment:** Engage users in interactive games, quizzes, or storytelling activities, making it fun for children and adults.
- 9. Personal Assistant:** Handle daily tasks such as setting reminders, controlling other smart devices, or providing weather updates.
- 10. Marketing and Advertising:** Act as an attention-grabbing tool in public spaces, promoting products or services with interactive voice communication.

1.6 Project Planning and bill of materials

1.6.1 Project Planning

1. Project Overview: The WiFi Talking Robot will be a robot that can move in all directions, respond to voice commands, and interact using Google Assistant (voice input and output). The robot will use the ESP8266 to connect to the internet and communicate with Google Assistant. It will also include hardware components like motors, Bluetooth, and an amplifier to ensure mobility and voice interaction.

2. Project Requirements

*** Hardware**

- ESP8266 (for WiFi connectivity)
- Relay Module (to control high-power components like motors)
- Step-up Transformer (for converting 5V to 15V for the motors)
- 15V Battery (to power the motors)
- Amplifier (for voice output)
- Bluetooth Module (for communication between ESP8266 and mobile device)
- 4 DC Motors (for movement)
- 4 Wheels (for robot mobility)
- Chassis (to hold all components together)
- Microphone and Speaker (for voice input and output)

*** Software**

- Google Assistant SDK (for voice control)
- Bluetooth Communication Protocol (for communication between ESP8266 and phone)
- Motor Controller Library (to control DC motors)

3. Milestones and Timeline

***Week 1: Research and Component Procurement**

- Research on available components and order all hardware components.
- Study the working of ESP8266 with Google Assistant and Bluetooth communication.

***Week 2: Circuit Design and Assembly**

- Design the schematic and wire the components, ensuring correct power distribution (motors and relay controlled by the ESP8266).
- Assemble motors, wheels, and chassis together.

***Week 3: Software Development**

- Write code to connect ESP8266 to the Wi-Fi network and communicate with the Google Assistant API.
- Develop code for motor control via Bluetooth (use Bluetooth commands for movement).

***Week 4: Integration of Components**

- Integrate Bluetooth communication, voice recognition, and motor control functionalities.
- Test communication between Google Assistant and the robot.

***Week 5: Final Testing and Debugging**

- Perform final testing on robot's mobility (movement in all directions).
- Test the voice command feature for correct responses.
- Debug issues with movement or connectivity.

*Week 6: Documentation and Presentation

- Prepare documentation on circuit design, software used, and results.
- Create a project report and presentation.

4. Tasks Breakdown

*Hardware Setup

- Assembly of Motors: Attach the DC motors to the chassis and wheels.
- Power Supply: Connect the step-up transformer and battery for motor power.
- Relay Wiring: Wire the relay to control the high-power motors from the ESP8266.

Software Development

- ESP8266 Setup: Program the ESP8266 to connect to Wi-Fi and handle Bluetooth commands.
- Google Assistant Integration*: Set up Google Assistant on the mobile device to accept voice commands.
- Bluetooth Communication*: Write code for Bluetooth communication to receive commands from the phone and send them to the ESP8266.
- Motor Control: Program the robot to move forward, backward, left, and right based on Bluetooth commands.
- Voice Interaction
 - .Voice Input: Set up microphone to receive voice input.
 - .Voice Output: Use an amplifier and speaker to deliver voice responses.

5. Risk Management

- Power Supply Issues: Ensure proper voltage regulation and check if the battery can supply enough power for the motors.
- Connectivity Issues: Test Wi-Fi and Bluetooth connections in various environments.
- Voice Recognition: Ensure that the Google Assistant responds correctly to voice commands in various noise conditions.

6. Future Enhancements

- Add autonomous navigation with sensors (ultrasonic or IR sensors).
- Implement a custom voice assistant for offline voice control.
- Use a mobile app to control the robot in addition to voice commands.

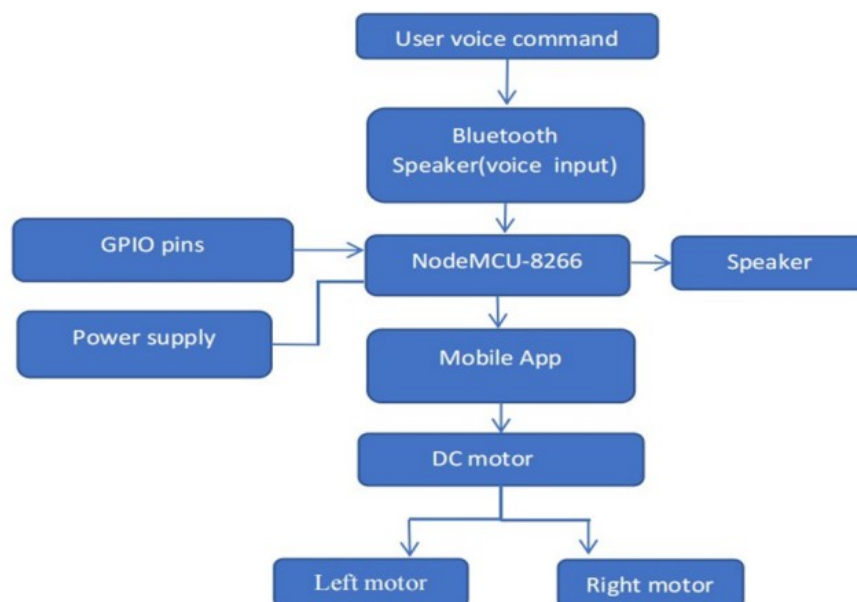
1.6.2 Bill of Materials

1. ESP 82266 - 350
2. 18650 cells - $2 \times 60 = 120$
3. Cell holder - 30
4. Switches - $2 \times 10 = 20$
5. Relay module - 360
6. Step down module - 60
7. Bluetooth module - 120
8. Amplifier module - 70
9. Breadboard - 80
10. Jumper wires - 50

Chapter 2

System design

2.1 Functional block diagram



2.2 Block diagram Explanation

1. Power Supply Block: Provides the necessary voltage and current to all components in the system. A 15V battery and a step-up transformer ensure stable power for high-demand components like motors and the amplifier.

2. Microcontroller (ESP8266) The brain of the system that controls all operations. Handles WiFi connectivity, voice communication, and motor control. Receives voice commands via the internet (Google Assistant) and processes them.

- 3. WiFi Module:** Integrated with the ESP8266, this enables the robot to connect to the internet. Facilitates communication with Google Assistant for voice-based queries and responses.
- 4. Motor Driver/Relay Control:** Interface between the microcontroller and the DC motors. Converts low-power signals from the ESP8266 into high-power signals to drive the motors. Relays or motor drivers control the direction and speed of the robot's movement.
- 5. DC Motors and Wheels:** Provide mobility to the robot, allowing it to move in all directions. Controlled by the motor driver or relay system.
- 6. Voice Input (Google Assistant):** Users interact with the robot through voice commands. The ESP8266 processes these commands via Google Assistant and generates appropriate responses.
- 7. Voice Output (Amplifier and Speaker):** Converts processed voice data into audible responses. The amplifier boosts the signal strength for clear sound output through the speaker.
- 8. Bluetooth Module:** Provides an alternative communication method, allowing manual control of the robot using a smartphone or other Bluetooth-enabled devices.
- 9. Step-Up Transformer:** Increases the battery voltage when required, ensuring that components like the motors and amplifier receive sufficient power.
- 10. Battery (15V):** The primary power source for the robot, supplying energy to the microcontroller, motors, and amplifier system.

Working Overview

- 1. Power Supply:** The 15V battery powers the entire system, with the step-up transformer managing power demands for high-energy components.
- 2. Input Command:** The user gives a voice command via Google Assistant, transmitted to the ESP8266 through WiFi.
- 3. Processing:** The ESP8266 processes the command, determines the appropriate response, and sends control signals to the motor driver or voice output system.
- 4. Mobility:** The motor driver or relays control the DC motors to move the robot in the desired direction.
- 5. Voice Response:** The robot's response is amplified and played through the speaker for user interaction.
- 6. Bluetooth Control:** Users can alternatively control the robot manually using a Bluetooth-connected device.

2.3 Design alternatives

1. Microcontroller Choice

*Alternative : Use ESP32

- Pros: Dual-core processor, more GPIO pins, integrated Bluetooth.
- Cons: Slightly more expensive.

2. Motor Control Mechanism

*Alternative: Motor Driver IC (e.g., L298N)

- Pros: Compact, faster switching, supports PWM for speed control.
- Cons: Slightly more complex wiring.

3. Voice Input Processing

*Alternative: Offline Voice Processing (e.g., PocketSphinx)

- Pros: Works offline, faster response.
- Cons: Limited vocabulary, requires more processing power.

4. Wireless Communication

Alternative: Bluetooth for Voice Commands

- Pros: Local communication, no need for internet.
- Cons: Limited range, less suitable for cloud-based services.

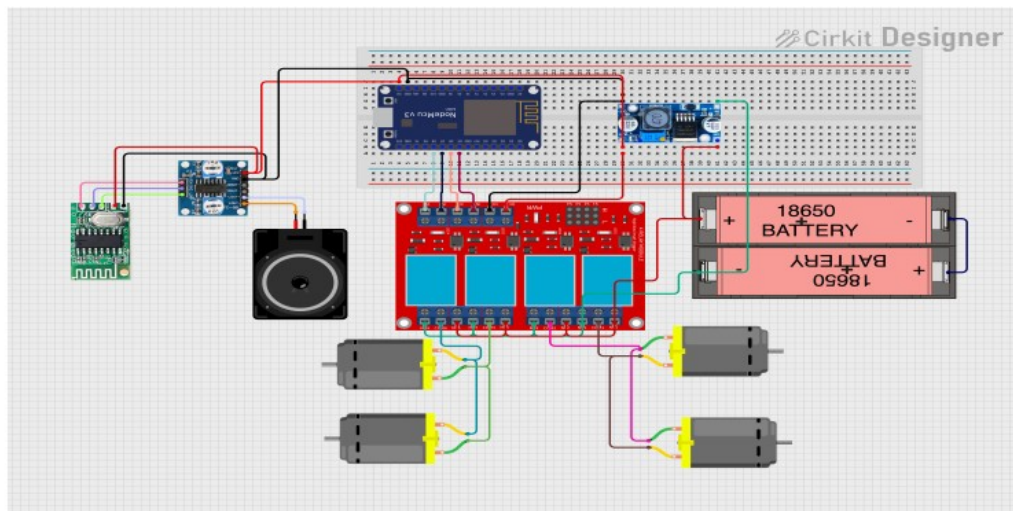
5. Audio Output

*Alternative: Direct Output to Small Buzzer

- Pros: Simple and compact.
- Cons: Poor sound quality, limited functionality.

Chapter 3

Implementation details



- ESP8266: Connects to Wi-Fi and integrates with Google Assistant for voice commands.
- Motor Driver and DC Motors: Control movement in all directions (forward, backward, left, right).
- Amplifier and Speaker: Provide clear voice output for responses.
- Power Supply: 15V battery with a relay and step-up transformer for efficient power management.
- Remote Control: Operated via a mobile app or Bluetooth for additional functionality.
- Voice Interaction: Uses speech-to-text conversion to process commands and respond with real-time audio.

3.1 Specifications

NodeMCU

- NodeMCU is predicated on the Espressif ESP8266-12E Wi-Fi System-On-Chip. It is based on Lua-based firmware and is open-source.
- The ESP8266 is more updated and younger than Arduino, and therefore the ESP has stronger specifications than Arduino.
- Operating Voltage: 2.5 to 3.3V
- Operating current: 800 mA
- 3.3V 600mA on-board voltage regulation

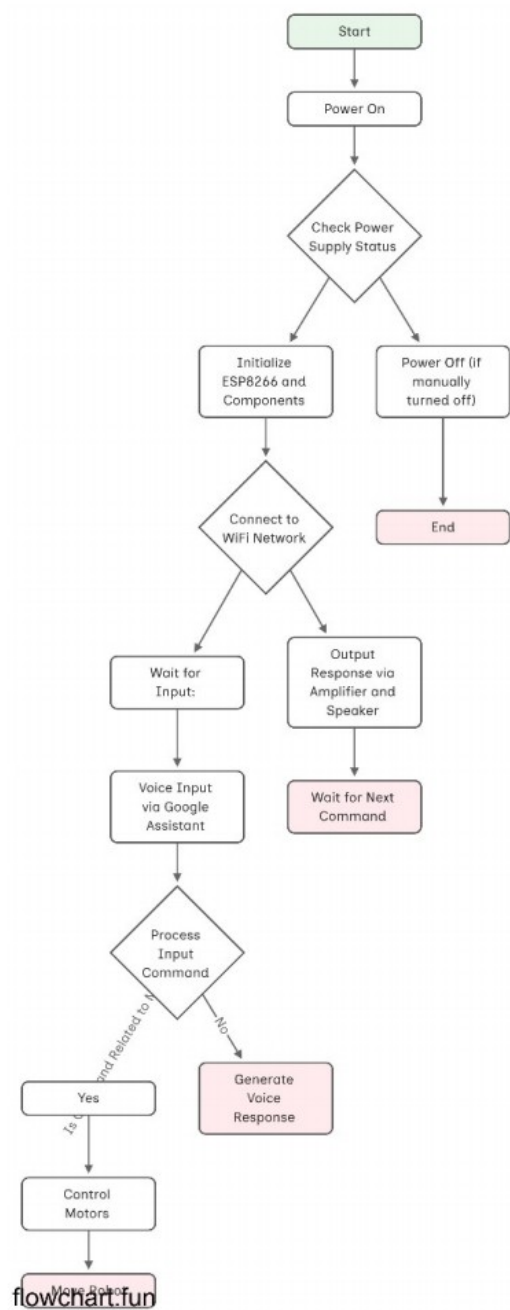
Relay module

- Normal voltage: 5V DC
- Normal current: 70mA
- Maximum load current: 10A/250V AC, 10A/30V DC
- Maximum switch voltage: 250V AC, 30V DC
- Operate time: 10ms

Amplifier - PAM8403

- 2 channels 3 W PAM8403 audio amplifier
- Output Power: 3 W + 3 W (at 4 ohm)
- Working Voltage: 2.5 to 5.5 V
- Board Size: 24 x 15 mm
- High amplification efficiency 85

3.2 Flowchart



Chapter 4

Optimization

4.1 Introduction to optimization

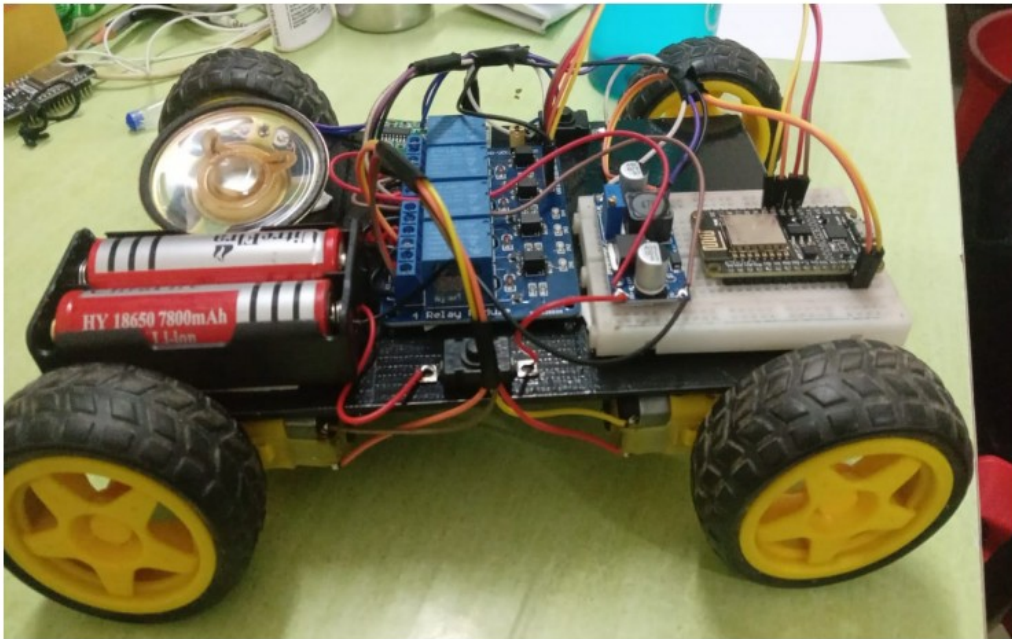
Optimization in engineering and technology is the systematic process of refining systems, algorithms, or processes to enhance their efficiency, performance, and reliability. It involves analyzing the current state of a system, identifying areas for improvement, and implementing changes to achieve better outcomes. By focusing on reducing waste, minimizing resource consumption, and improving functionality, optimization aims to streamline operations and maximize output. This continual pursuit of efficiency not only enhances competitiveness and innovation but also ensures that technological solutions are sustainable, scalable, and resilient to meet evolving demands in today's dynamic landscape.

4.2 Optimization of the project

- 1) Energy Efficiency: Use ESP8266 deep sleep and PWM for power saving.
- 2) Battery Management: Add a BMS and voltage monitoring via ESP8266 ADC.
- 3) Power Supply: Add a voltage regulator and capacitors for stable voltage.
- 4) Feedback System: Enable audio feedback for commands via the speaker.

Chapter 5

Results



Chapter 6

Conclusions and future scope

6.1 Conclusion

The WiFi-enabled talking robot is a promising integration of IoT, robotics, and AI, demonstrating the potential of voice-interactive systems in everyday life. By combining components like the ESP8266 module, DC motors, and a voice assistant, the project creates a robot capable of providing information, responding to voice commands, and moving autonomously.

This project serves as an excellent platform for learning and exploring new technologies. The robot's applications span education, healthcare, retail, and entertainment, showcasing its versatility. With further optimization and advancements, the robot can evolve into a more sophisticated system with broader capabilities, paving the way for more impactful innovations in robotics and IoT.

6.2 Future scope

1. **Obstacle Detection and Avoidance:** Integrating sensors like ultrasonic or IR for autonomous navigation.
2. **Improved AI:** Using more advanced AI models for better understanding and contextual responses.
3. **Battery Optimization:** Employing efficient power management systems for longer operation times.
4. **Video Streaming:** Adding a camera module for real-time video surveillance and interaction.
5. **Integration with Smart Home Systems:** Enabling control of other smart devices.
6. **Enhanced Mobility:** Using more advanced motors or wheels for rough terrains.

Chapter 7

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